

CRISTIAN TURCAN

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EDUCATION

Delft University of Technology

MSc Data Science & Artificial Intelligence Technology

BSc (Honours) Computer Science & Engineering

Minor in Finance

Delft, Netherlands

Sep. 2026 – Jun. 2028

Sep. 2023 – Jun. 2026

Sep. 2025 – Jan. 2026

EXPERIENCE

Software Engineer (Part-time)

Offroad Apps (VitalsIQ)

Jan. 2026 – Present

Delft, Netherlands

- Built a .NET 10 Android binding of Garmin's native Health SDK (Standard SDK 4.7.0) and a .NET MAUI companion app that pairs Garmin wearables and streams 7 live vital-sign types (heart rate, HRV, SpO₂, respiration, stress, steps, accelerometer) to the VitalsIQ backend.
- Authored the Java-to-C# binding transforms resolving R8, Guava, SQLCipher, and Gson interop conflicts, plus an Android foreground service for resilient background sync.
- Built a CivTAK (civilian ATAK) plugin in Java that surfaces live wearable vitals as an in-app pane and map marker, on a device-independent, JUnit-tested core with a bound-service Messenger IPC bridge to the MAUI app.
- Ported the full JABCode color 2D-barcode library (encoder and decoder with error correction) from C to C#.

Embedded Software Engineer (Part-time)

Insyght.io

Aug. 2025 – Present

Delft, Netherlands

- Built the end-to-end noise-classifier model pipeline (data preparation, training, evaluation, and firmware integration) for on-device inference on constrained hardware; the classifier achieved 92% accuracy at a size of only 40KB.
- Developed *AuraSense* firmware in C/C++ on Zephyr RTOS / nRF Connect SDK for a solar-powered nRF54L15 + nRF9151 noise and air-quality IoT monitor; architecting 11 self-contained driver components (~3K LOC).
- Built a real-time acoustic sound-level-metering pipeline: A-weighted (3-stage biquad IIR) dBA and Leq over 10s windows from stereo 16 kHz PDM MEMS microphones, with calibrated dB-SPL and clipping detection.
- Cut measured idle current from ~960 μ A to ~189 μ A via PPK2 differential power profiling, root-causing the LTE-modem floor, an LDO-vs-DC/DC path, and a mis-wired PMIC GPIO.
- Implemented an LTE-M/NB-IoT uplink (nRF9151 serial modem, AT sockets, hand-written JSON codec) to a Node.js/Express telemetry backend deployed on Railway.

Machine Learning Engineer Intern

Altix Capital

Apr. 2025 – Jun. 2025

Amsterdam, Netherlands

- Built an AI document-ingestion pipeline to parse PDFs, spreadsheets, and images (with OCR), then auto-summarize and classify content. Reduced manual screening time of Analysts from 10 minutes per document to only 2 minutes.
- Shipped a REST service and lightweight web UI that streamlined analyst review, cutting document navigation and handling time by roughly 60%.

Machine Learning Engineer Intern

Huawei

Dec. 2024 – May 2025

Amsterdam, Netherlands

- Designed an English keyword-spotting pipeline. The precision went from 92% to 94% and reduced WER by 1.1%.
- Built, cleaned, and validated large-scale audio datasets with forced alignment and transfer learning. Expanded the coverage of audios from 100 to more than 300 hours and reduced the number of corrupted data from 10% to 2%.
- Built a reproducible benchmarking pipeline for ASR/TTS candidates. Cut evaluation time from 4 hours to 1 hour.

PROJECTS

Perception-Aware 3D Mesh Simplification | C++, Computational Geometry

- Built a from-scratch solver for Huawei's 2nd IMC Challenge: QEM edge-collapse with a best-first priority queue, plus custom software renderer, SSIM, and Hausdorff implementations to self-score against the hidden judge.
- Compressed closed-manifold meshes up to 1.1M vertices, achieving a judge score of 87.49 with all 7 test cases accepted, backed by a C++ unit-test suite.

CreatorMatch — AI Creator-Discovery Platform | Next.js, FastAPI, Postgres/pgvector

- Built a full-stack platform that matches businesses to local content creators: a FastAPI + Celery backend ranks creators with a weighted scoring formula and LLM brand-fit scoring over pgvector semantic search, served through a Next.js dashboard with auto-drafted outreach.

IMC Prosperity 4 — Algorithmic Trading | Python, NumPy, pandas, SciPy

- Implemented options market-making with Black-Scholes pricing and a quadratic implied-volatility smile fit, alongside mean-reversion on the underlying asset.
- Wrote a stationarity/time-series test library (ADF, KPSS, Ljung-Box, variance-ratio) and a grid-search optimizer, validated over a submodule backtester.

Telegram & Automation Bots | Python, python-telegram-bot, aiogram, Playwright

- Built a change-tracking Telegram bot that snapshot-diffs paginated third-party API data and alerts on deltas, with thread-safe SQLite persistence, retry/backoff with pagination, and 41 unit tests across 8 modules.

- Automated TU Delft gym-slot booking with a Playwright bot that drives a full OIDC/SAML NetID SSO login and retries on booking race conditions.
- Architected a broker-based microservice Telegram bot for local event aggregation: aiogram with Celery + Redis workers, Postgres with Alembic migrations, orchestrated via Docker Compose.

JIT Compiler for Brainfuck | Assembly

- Built a just-in-time compiler for the Brainfuck language entirely in assembly; a self-directed project driven by an interest in low-level programming and code generation.

AWARDS

AlgoTrade 2025 Hackathon	7 th Place	Croatia	Jun. 2025
2023 Post World Finals Online ICPC Challenge	4 th Prize	Online	May 2024
National Mathematical Olympiad	Bronze Prize	Republic of Moldova	Mar. 2023
National Mathematical Olympiad	Silver Prize	Republic of Moldova	Mar. 2022
National Informatics Olympiad “Game of Code”	Bronze Prize	Moldova State University	Dec. 2021

RESEARCH

Honours Program — Decision-Focused Learning

Sep. 2024 – Present

First Author, TU Delft

- First-author paper “*Learning Feasible Decisions Directly from Features for Combinatorial Optimization*” (under review at BNAIC 2026). Proposed LR-STE, a solver-free predict-then-optimize method that projects a neural predictor’s output onto the problem’s linear relaxation and applies a feasibility-correction layer, trained end-to-end with a straight-through estimator to minimize decision regret rather than prediction error.
- Benchmarked on 0/1 Knapsack and the Traveling Salesperson Problem against SPO+, two-stage predict-then-optimize, and projection-free (STE) baselines; on TSP, LR-STE reached the lowest regret among solver-free methods — second only to the far costlier SPO+ — while staying cheap to both train and run.
- Adapted an attention-based REINFORCE routing solver (Kool et al., ICLR 2019) to the decision-focused setting, and built a config-driven, unit-tested experiment framework (PyTorch, Gurobi, CVXPY/cvxpylayers, PyEPO, Optuna, WandB) with SLURM scale-out on TU Delft’s DelftBlue HPC cluster and CI.

Bachelor Research Project (CSE3000)

Apr. 2026 – Jun. 2026

Sole Author, TU Delft

- Sole-author paper “*Heterophilic Methods on Multi-Label Graphs.*” Benchmarked 8 models — MLP and DeepWalk baselines plus 6 heterophily GNNs (H2GCN, GPR-GNN, LINKX, FAGCN, ACM-GCN, Ordered GNN) — across 5 real and 2 synthetic multi-label graphs in PyTorch Geometric.
- Ran controlled homophily and label-cardinality sweeps over multiple seeds with fixed splits; found node-classification performance (Average Precision) tracks graph homophily more than model sophistication — heterophily GNNs rarely beat a feature-only MLP or structure-only DeepWalk in the low-homophily regime.
- Built *gnmbench*, a single config-driven pipeline with a synthetic multi-label graph generator and a model registry, hardened with split-leakage tests and parallelized as DelftBlue SLURM array jobs.

ACTIVITIES

Susquehanna International Group (SIG) — Open Days

Apr. 2025

Dublin, Ireland

- 3-day immersion at SIG, with training sessions on options pricing and market making.

Jane Street — Estimation

May 2025

Netherlands

- Competed in Jane Street’s team estimation competition alongside traders and quants.

SKILLS & INTERESTS

Languages: Python, C/C++, C#, Java, Haskell, SQL, Bash, JavaScript, TypeScript

ML & Data: PyTorch, PyTorch Geometric, TensorFlow, SciKit-Learn, Pandas, NumPy, SciPy, Matplotlib, XGBoost, Transformers, HuggingFace, Optuna, Gurobi, CVXPY

Frameworks & Systems: Zephyr RTOS, nRF Connect SDK, .NET / MAUI, FastAPI, SpringBoot, Next.js, React, Node.js, Celery, Redis, PostgreSQL

Developer Tools: Git, Docker, Linux, CUDA, Playwright, pytest, WandB, GitLab, VS Code, PyCharm, IntelliJ

Interests: Stock Investing, Cryptocurrency, Music, Perfumes, Taekwondo, Cooking (Best Alfredo Pasta), Board Games

Favorite Books: The Idiot, White Nights, Roadside Picnic, Monday Starts on Saturday, Norwegian Wood, The Little Prince

Favorite Movies: Inception, Whiplash, Oldboy, The Shawshank Redemption, Coco, Your Name, The Wolf of Wall Street